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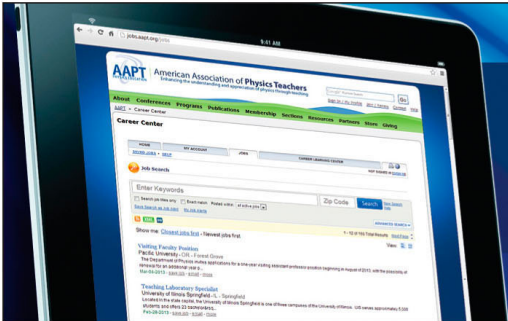
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An acoustic demonstration of an avoided crossing

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We experimentally demonstrate an avoided crossing in an acoustic system, consisting of two coupled PVC tube sections. One section has a fixed length, while the other has a variable length. Coupling between these tube sections is controlled by an aluminum diaphragm with a variable hole size. The avoided crossings in the acoustic system are compared to those of a quantum infinite square well split into two regions—one of fixed length and another of variable length. The two regions are separated by a delta potential well that controls the coupling between the two regions. We demonstrate that the acoustic and quantum systems exhibit similar avoided crossing behaviors. © 2017 American Association of Physics Teachers.

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I. INTRODUCTION

Avoided crossings are a classical phenomena, predicted in mechanical,¹ optical,² and electrical systems.³ However, students typically first encounter avoided crossings in a quantum mechanics setting, where there are many examples.^{4–6} In general, avoided crossings occur in coupled systems. Without coupling, if the two systems are made identical, they have the same eigenvalues and degenerate states. Coupling introduced between the systems can break this degeneracy. Each eigenvalue splits into two separate values in a process called level splitting. Level splitting no longer allows the eigenvalues to cross. An avoided crossing can be visualized by looking at measured eigenvalues when adjusting a parameter of the system (see Fig. 1). When the two systems are far from identical, coupling has a weak effect, making the measured eigenvalues for the coupled systems (solid line) similar to the eigenvalues when uncoupled (dashed line). As the parameter is tuned to make the systems closer to identical, the two eigenvalues approach each other. Once the adjustable parameter is tuned for the two systems to be nearly identical, the eigenvalues repel each other instead of crossing, creating an avoided crossing. Here, we demonstrate this phenomenon in an acoustic system and compare it to a particular quantum mechanical system.

A quantum system that is commonly studied in undergraduate physics⁷ is an infinite square well with a central delta potential well. This system has the potential energy

$$V(x) = \begin{cases} -\alpha\delta(x-L) & \text{for } 0 < x < 2L + \Delta L \\ \infty & \text{otherwise.} \end{cases} \quad (1)$$

To have a common vocabulary between this system and the experimental acoustic system discussed later, we define a “cavity” for the quantum system to be a region bounded by a wall of the infinite square well and the delta potential well. The “length” of a cavity is defined as the distance between the infinite square well wall and the delta potential well. The potential energy of this system is depicted in Fig. 2(a). One cavity has fixed length L , while the other contains a movable wall, allowing for an adjustable length of $L + \Delta L$. The delta potential at $x = L$ controls the coupling between the two cavities based on the delta potential well “strength” α . When α

approaches infinity, the two cavities are uncoupled. When uncoupled, each cavity has eigenenergies E_n given by

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}, \quad (2)$$

where a is the width of the cavity. As the value of α decreases, a particle in one cavity is more likely to traverse through the delta-potential well to the other side, increasing the coupling between the cavities. This coupling affects the eigenvalues of the system.

This quantum system can be compared to an acoustic system composed of two PVC pipe sections coupled by an aluminum diaphragm with a specific hole size. A schematic of the system is depicted in Fig. 2(b). One pipe section, or cavity, is fixed at length L , while the other has a movable wall, allowing for an adjustable length of $L + \Delta L$. The hole in the aluminum diaphragm controls the coupling between the two cavities. When the hole diameter is zero (the diaphragm is a solid disk), the two cavities are essentially uncoupled. When uncoupled, each cavity supports resonance frequencies f_n given by

$$f_n = \frac{nv}{2a}, \quad (3)$$

where v is the speed of sound in air and a is the length of the cavity. As the hole diameter increases, a larger amplitude of a sound wave incident on the disk traverses the disk to the other side, increasing the coupling between the cavities. This coupling affects the resonance frequencies of the system.

The acoustic system has already been shown to experimentally follow the same level splitting⁸ and band structure⁹ predicted for the quantum mechanical system. These two systems are comparable because quantum mechanics and acoustics follow the same wave equation in time-independent situations. The quantum system is governed by the time-independent Schrödinger equation

$$\frac{\partial^2\psi}{\partial x^2} = k^2\psi, \quad (4)$$

where $k = \sqrt{2mE/\hbar^2}$, because the potential energy within the cavities is zero. A pressure wave p is governed by the acoustic wave equation

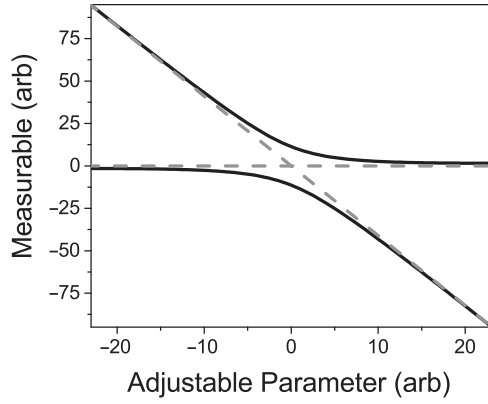


Fig. 1. Example plot that demonstrates an avoided crossing in a system containing two subsystems. The two subsystems are identical when the adjustable parameter is zero. Dashed lines show the results when no coupling exists between the subsystems; solid lines show the results with coupling. Without coupling, the measured values of the subsystems can cross. Introducing coupling when the subsystems are close to identical causes the measured values to repel away from each other, forming an avoided crossing.

$$\frac{\partial^2 p}{\partial t^2} = \frac{1}{v^2} \frac{\partial^2 p}{\partial x^2}, \quad (5)$$

where $v = \sqrt{B/\rho_0}$ is the speed of sound in the medium with bulk modulus B and equilibrium density ρ_0 . In time independent situations, such as standing waves, this reduces to

$$\frac{\partial^2 p}{\partial x^2} = k_s^2 p, \quad (6)$$

where $k_s = 2\pi f/v$ with f being the sound wave frequency. Equations (4) and (6) show that the two governing equations have the same mathematical form.

This congruence between the two systems suggests they will demonstrate similar avoided crossings, because both systems are being analyzed under time-independent conditions. Section II of this paper develops the theory behind the avoided crossing of the quantum system, and Sec. III

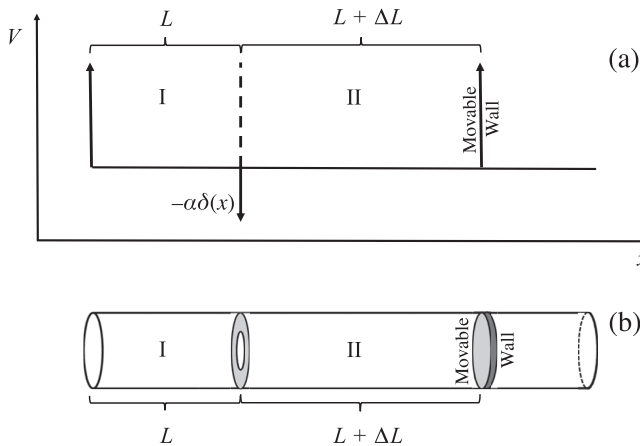


Fig. 2. (a) The quantum system consists of an infinite square well of length $2L + \Delta L$ with a delta well potential of strength α at $x = L$. The left cavity (I) has a fixed length L while the right cavity (II) has a variable length of $L + \Delta L$. (b) A similar acoustic system is a pair of PVC pipe sections separated by an aluminum diaphragm with an adjustable hole. The left pipe section has a fixed length L while the right pipe section has a variable length of $L + \Delta L$.

describes the experimental acoustic system. Section IV demonstrates the avoided crossings for the quantum and acoustic systems and compares them and analyzes the effect of coupling on the shape of the avoided crossings for both systems.

II. ANALYSIS OF THE QUANTUM SYSTEM

When the two cavities within the infinite square well are uncoupled ($\alpha \rightarrow \infty$), each region behaves as an isolated infinite square well of width a (see the Appendix for a more-detailed analysis). When the variable-length cavity has a different length than the static region, two distinct sets of eigenenergies are formed via Eq. (2). When the variable-length cavity matches the static cavity, they both have the same set of eigenenergies, so only one degenerate set is present.

To analytically determine the eigenenergies when the cavities are coupled, the Time-Independent Schrödinger Equation (TISE) must be solved for the potential in Eq. (1) with the appropriate boundary conditions. Because the potential is zero in both cavities, the general solutions for the wave functions, ψ_I and ψ_{II} , in the two cavities are plane wave solutions

$$\psi_I(x) = Ae^{-ikx} + Be^{ikx} \quad (7)$$

and

$$\psi_{II}(x) = Ce^{-ikx} + De^{ikx}, \quad (8)$$

where the subscript I corresponds to the static-length cavity and II refers to the variable-length cavity, as shown in Fig. 2(a), with A , B , C , and D being constants of integration. Since a particle in this system is constrained within the well, the wave function must be zero outside of the infinite square well.

Solutions to the TISE are found by applying the typical boundary conditions: continuity of the wave function at $x = 0$,

$$\psi_I(0) = 0; \quad (9)$$

continuity of the wave function at $x = 2L + \Delta L$,

$$\psi_{II}(2L + \Delta L) = 0; \quad (10)$$

continuity of the wave function at the delta potential well,

$$\psi_I(L) = \psi_{II}(L); \quad (11)$$

and discontinuity of the wave function's derivative at the delta potential well,¹⁰

$$\left. \frac{d\psi_{II}}{dx} \right|_{x=L} - \left. \frac{d\psi_I}{dx} \right|_{x=L} = -\frac{2m\alpha}{\hbar^2} \psi(L). \quad (12)$$

This last boundary condition comes from the derivative of a wave function potentially being discontinuous at locations where the potential energy is infinite. This discontinuity is found by integrating the TISE over a small region around the delta well potential.

These boundary conditions, combined with the general solutions for the wave functions, yield the transcendental equation

$$\cot(kL) + \cot[k(L + \Delta L)] = \frac{2m\alpha}{\hbar^2 k}, \quad (13)$$

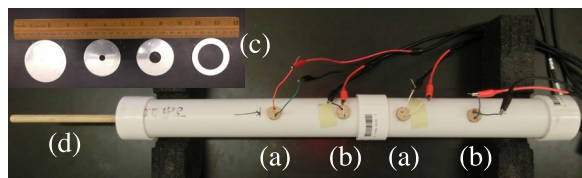


Fig. 3. The experimental apparatus uses two PVC tube sections. A sinusoidal signal is emitted within the tubular array by speakers (a). Microphones within the tube (b) pick up the sound signal. The cavities are separated by aluminum discs (c) of various hole diameters (0 mm, 8 mm, 23 mm, and 36 mm) to control the coupling between cavities. The length of the left cavity is controlled by a plunger (d).

for cases when $\Delta L \neq 0$. For the case where $\Delta L = 0$, the boundary conditions are satisfied when

$$\cot(kL) = \frac{m\alpha}{\hbar^2 k}. \quad (14)$$

A Maxima¹¹ program is used to numerically solve Eqs. (13) and (14) to find the wavenumbers associated with eigenenergies for different values of ΔL and α .

These quantum wavenumbers are comparable to the wavenumbers associated with resonance frequencies in the acoustic system. However, wavenumbers are not very intuitive in the acoustic system, whereas frequencies are. We define an eigenfrequency f of the quantum system to be the acoustic frequency associated with a particular wavenumber by

$$f = \frac{kv}{2\pi}, \quad (15)$$

where v is the speed of sound in air (which depends on the air pressure and temperature).

III. ACOUSTIC SYSTEM

The acoustic apparatus is comprised of two PVC tube sections, and is shown in Fig. 3. One PVC tube section is fixed at length $L = 0.27$ m. The second PVC tube section has a length of 0.36 m, but the endcap for this cavity is connected to a plunger [Fig. 3(d)]. Pushing the plunger in or out controls the variable cavity length. The two cavities are separated by a thin, aluminum diaphragm that can contain various hole diameters [Fig. 3(c)]: 0 mm (a solid disk), 8 mm, 23 mm, and 36 mm. The endcaps for the static cavity and the variable cavity are both plastic cylinders with a thickness of 5 mm.

Resonance frequencies are found by using frequency sweeps within the tubular array. A function generator¹² sends a sinusoidal signal to both speakers,¹³ one inside each tube section. The function generator makes a linear sweep over a frequency range of 1300–3350 Hz with a sweep rate of 8.2 Hz/s. This sweep rate, in conjunction with the leaking of the sound from the tubes, results in a 10-mHz range of frequencies present in the tube at any given time. The sound in the tube is picked up by a microphone¹⁴ in each tube section. The sinusoidal signal from the microphones is amplified¹⁵ and rectified to provide a dc signal. The rectified signal is then sent to the data acquisition board.¹⁶ The function generator has a second output, a “Sweep” signal, which is a voltage proportional to the frequency being emitted. This “Sweep” signal is also picked up by the data acquisition board and is used to convert time to frequency by making use of the sweep rate.

Example spectra like those found in Fig. 4 allow identification of resonance frequencies. These are the frequencies that produce local maxima in amplitude. Setting the variable cavity to the same length as the static cavity with no

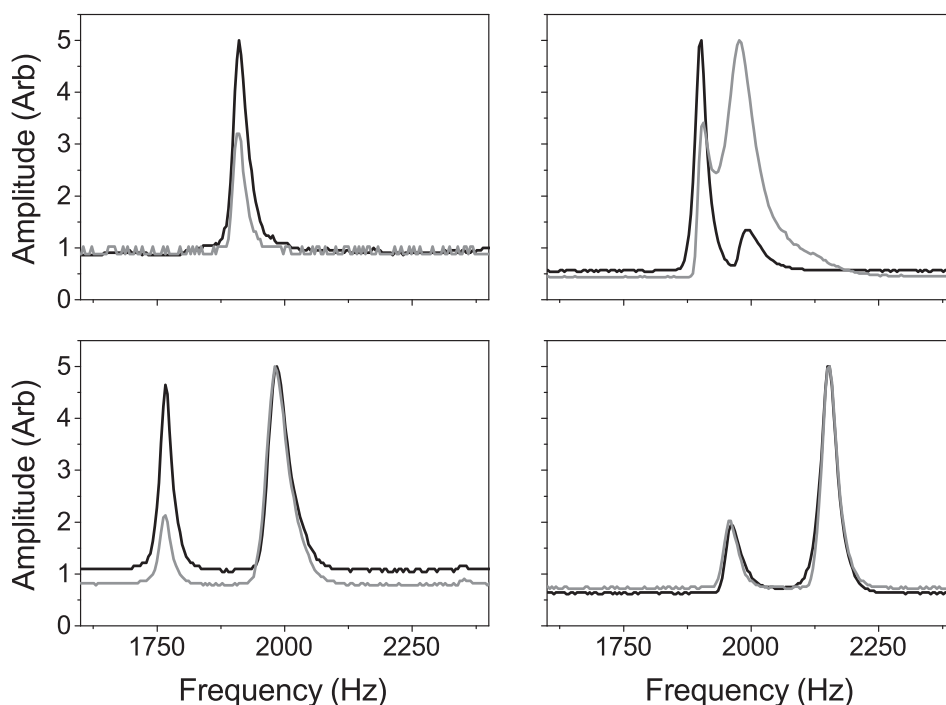


Fig. 4. Resonance frequencies for the acoustic system are found by analyzing spectra from frequency sweeps. Each cavity has its own microphone (gray for static cavity and black for variable cavity). Example spectra are shown with both cavities having the same length (270 mm) and no coupling (top left), both cavities having the same length with an 8-mm hole connecting them (top right), the variable cavity 25 mm longer with an 8-mm hole (bottom left), and the variable cavity 25 mm shorter with an 8-mm hole (bottom right).

coupling between the two cavities yields a single resonance (top left). Adding coupling to this system results in a pair of resonances (top right), demonstrating level-splitting. Setting a cavity-length difference of $\Delta L = 25$ mm shifts the peaks to lower frequencies (bottom left), while a cavity-length difference of $\Delta L = -25$ mm shifts the peaks to higher frequencies (bottom right). Spectra were taken from both cavities simultaneously; the gray curve is from the static cavity while the black curve is from the variable cavity. These curves are scaled in the y-axis to ease comparisons between cavity measurements.

The end caps of the system are not perfectly reflective, causing the resonances in the sweeps to have a finite width. This contrasts to the quantum system where the walls of the infinite square well are assumed to be perfectly reflective, yielding discrete eigenvalues. When the solid diaphragm connects the two cavities [upper left panel of Fig. 4], it does not perfectly prevent coupling because it is not perfectly reflective. However, the resulting coupling is so weak that any level splitting that occurs is much smaller than the resonance width. Experimentally, this means that such level splitting is undetectable in the frequency sweep and is practically non-existent.

The differences in amplitude of the resulting peaks can be attributed to microphone placement within the tube. A microphone located near a node of the sinusoidal wave will result in a low response, regardless of the amplitude of the standing wave. A microphone located near an antinode of the sinusoidal wave will result in a higher response. Microphones are intentionally placed at the different positions in the cavities to ensure no resonances are missed; however, this causes amplitude of a resonance to not necessarily

be the same in both cavities. The peak amplitudes are unimportant; the frequencies associated with the peaks are the focus. Both cavities give the same resonance frequencies for any variable cavity length and non-zero coupling (i.e., when the diaphragm has a non-zero hole size).

IV. RESULTS

The wave numbers corresponding to eigenstates of the quantum system are found for four different values of delta potential strength: $\alpha = (3 \times 10^{10}) \hbar^2/m$, $100 \hbar^2/m$, $30 \hbar^2/m$, and $0.3 \hbar^2/m$. These values are chosen to show the full spread of effects on the avoided crossing by the value of α . We determine the eigenfrequencies at $n = 3$ for each value of α to match the frequencies used in the acoustic system. The fundamental and first harmonic of the acoustic system get muddled due to the response of the microphone and speaker.

For large α (Fig. 5, top left) the coupling between the two cavities is essentially zero, so a crossing of eigenfrequencies is apparent. In this case, each eigenfrequency in the pair can be approximately assigned to a particular cavity. The set of eigenfrequencies that make the flat, horizontal line can be attributed to the static-length cavity. These eigenfrequencies always have approximately the same value for any length of the variable-width region. The set that follows the non-horizontal curve can be associated with the variable-length cavity, which consists of a continually changing eigenfrequency that depends on the cavity length via Eq. (2).

Reducing the delta potential strength allows more coupling between the cavities and results in an avoided crossing (Fig. 5, top right). Note that these eigenstates are no longer associated with a particular cavity; both sets are eigenstates

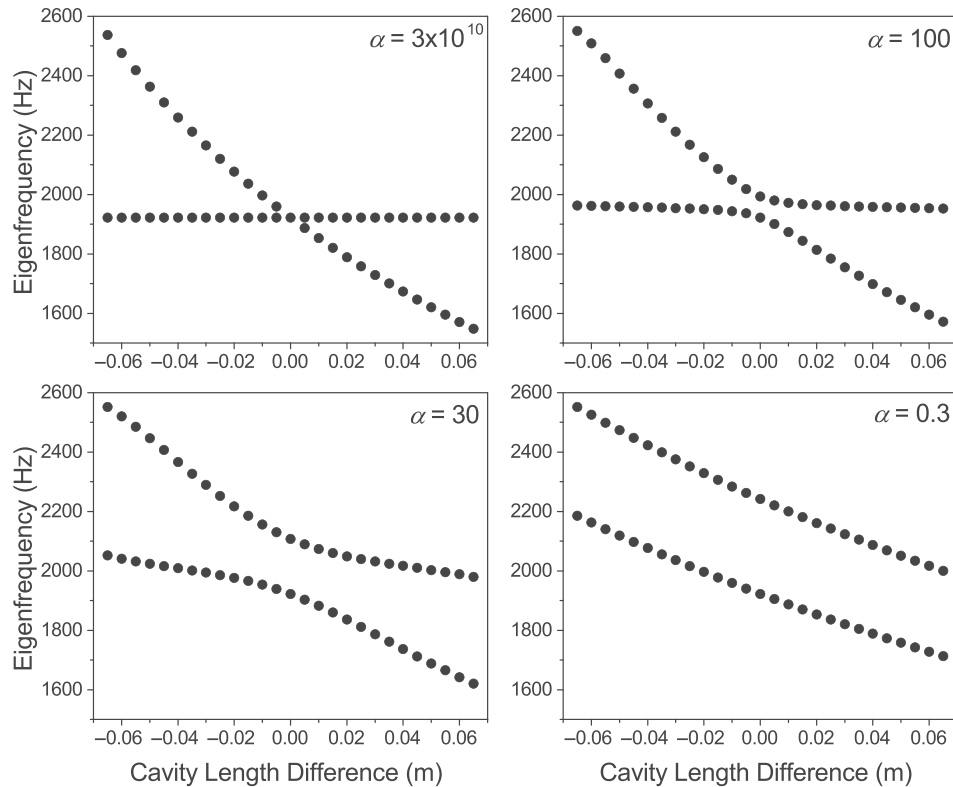


Fig. 5. Theoretical plots of the eigenfrequency dependence on cavity-length difference demonstrate the emergence of avoided crossings in the quantum system. A large delta-potential strength [$\alpha = (3 \times 10^{10}) \hbar^2/m$, top left] allows a crossing. Reducing the delta-potential strength [$\alpha = 100 \hbar^2/m$, top right; $\alpha = 30 \hbar^2/m$, bottom left] yields avoided crossings. Too much coupling [$\alpha = 0.3 \hbar^2/m$, bottom right] washes out the avoided crossing.

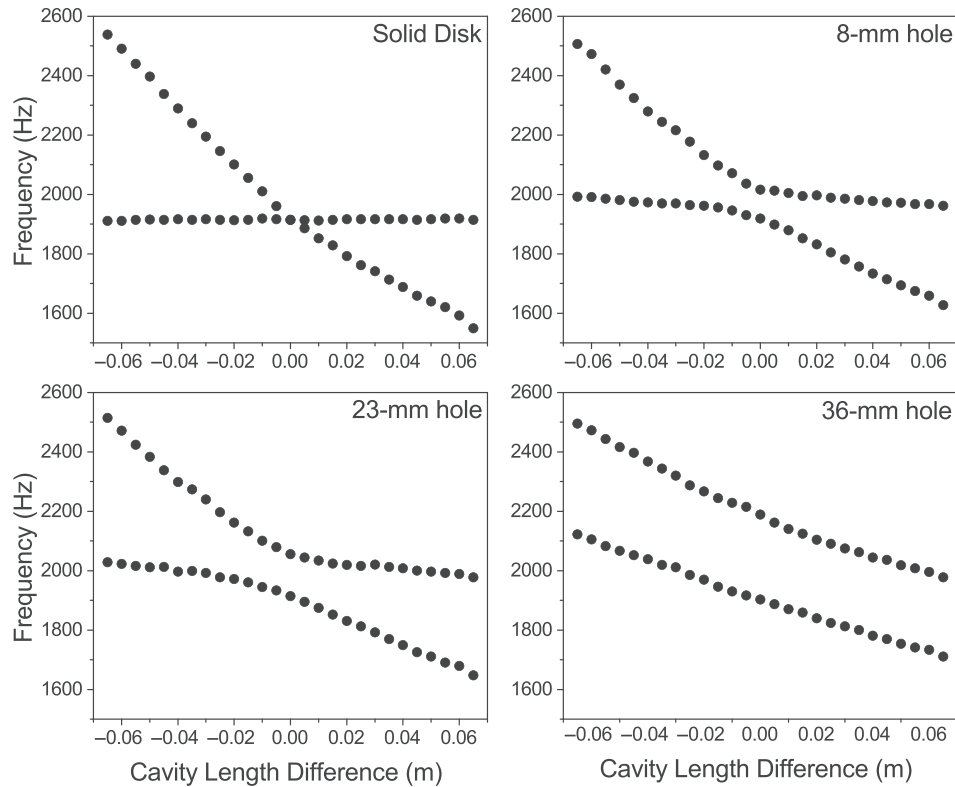


Fig. 6. Experimental plots of the resonance frequency dependence on cavity-length difference demonstrate the emergence of avoided crossings in the acoustic system. A solid diaphragm uncouples the two tube sections (top left), which yields a crossing. When coupling is introduced with a 8-mm diameter hole (top right) or a 23-mm hole (bottom left) the avoided crossing becomes visible. Using a 36-mm diameter hole (bottom right) adds too much coupling, washing out the avoided crossing.

of the entire system. When the cavity-length difference is very large, eigenfrequencies are similar to those in the crossing case. However, as $\Delta L \rightarrow 0$ the eigenfrequencies do not cross—this is an avoided crossing.

Reducing the delta-potential strength further increases the coupling, and causes a broader avoided crossing (Fig. 5, bottom left). The eigenfrequencies are further repelled at small cavity-length differences. At large cavity-length differences, more curvature is seen in the graph, indicating that the asymptotic behavior of the eigenfrequencies begins to dissipate.

A very small delta potential strength (Fig. 5, bottom right) washes out the avoided crossing to the point where it is no longer visible. The spacing between the two eigenfrequencies is essentially the same for all region-width differences. This closely matches two consecutive eigenfrequencies for an infinite square well of length $2L + \Delta L$.

A similar approach is taken for the acoustic system. The resonance frequencies are found for various diaphragm hole diameters, which control the coupling between the cavities. A solid diaphragm (Fig. 6, top left) effectively prevents coupling, just like in the quantum system with $\alpha = (3 \times 10^{10}) \hbar^2/m$. Here, we again see one set of resonance frequencies that are constant with changes to the cavity length difference, attributed to the static cavity. The second set of resonances are affected by the length L of the variable cavity and follow the expected resonance frequencies for a closed tube given in Eq. (3) with $n=3$. When the two cavities are the same length, they both have the same resonant frequency, which leads to a crossing of the resonance frequencies.

A small (8-mm) hole in the diaphragm allows the cavities to be slightly coupled (Fig. 6, top right). This situation gives

rise to an avoided crossing, just like in the quantum system with $\alpha = 100 \hbar^2/m$. A larger (23-mm) hole diameter further couples the cavities (Fig. 6, bottom left) and broadens the avoided crossing, mimicking the quantum system with $\alpha = 30 \hbar^2/m$. An even larger (36-mm) hole spaces out the resonances so that the difference between the frequencies is essentially constant for all variable-cavity lengths (Fig. 6, bottom right), much like $\alpha = 0.3 \hbar^2/m$ in the quantum system.

While we compare results for specific diaphragm-hole diameter with specific values of delta-potential strength (α), there is no direct relationship between these parameters that is independent of wave number. Because of this missing connection, we qualitatively chose values of α that gave results that resemble those from the acoustic system. Despite no one-to-one relationship between the couplings in the acoustic and quantum systems, both systems exhibit avoided crossings and demonstrate the same responses to increased coupling.

V. CONCLUSION

We have demonstrated an avoided crossing in an acoustic system and showed that they mimic avoided crossings in a simple quantum system. The similarities between the two systems make the pair a potentially valuable learning tool as a demonstration in a quantum mechanics course. The equipment required for the acoustic experiment is accessible to undergraduate students and typically available in an undergraduate laboratory, so this experiment could be adapted for an undergraduate advanced laboratory course.

It might be interesting to extend this idea of avoided crossing to a larger number of cavities. Changing the length of

one cavity within a large array would be a way to explore array defects, which has analogs to work done in condensed matter and crystalline structure.¹⁷ This system also creates closely spaced resonances with a tunable frequency difference that could be useful for various applications.

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APPENDIX: SOLUTIONS TO A REGION BOUNDED BY A POSITIVE INFINITE POTENTIAL AND AN INFINITE-STRENGTH DELTA POTENTIAL WELL

The transcendental equation for determining the wave numbers in the variable-width infinite square well with a delta well potential is Eq. (13) when $\Delta L \neq 0$. As $\alpha \rightarrow \infty$, the right side of Eq. (13) also goes to infinity. Solutions to this equation force either $\cot[kL]$ or $\cot[k(L + \Delta L)]$ to go to infinity, giving two sets of solutions,

$$k = \frac{n\pi}{L} \quad (\text{A1})$$

and

$$k = \frac{n\pi}{L + \Delta L}. \quad (\text{A2})$$

These two sets of solutions match the solutions of two infinite square wells, one of width L and the other of width $L + \Delta L$.

When ΔL is zero, solutions come from Eq. (14). As $\alpha \rightarrow \infty$, $\cot[kL]$ must also go to infinity, which gives the solutions as Eq. (A1). Thus, only one set of solutions, with two-fold degeneracy, occur when the two regions are the same width.

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